Individual Reflection: Customers Like You
Name: Adithya Anand

Before this project, I thought most of the work would be in building the models. I was wrong. A big part of my time went into just getting the data ready. The Online Retail II dataset had cancelled orders, wrong prices and quantities, and a lot of purchases with no customer ID. At first it felt like I was throwing away more data than I was keeping, but I realised that the models are only as good as what you feed them. Turning the cleaned data into a customer-product matrix was also new to me. I hadn't thought about how empty it would be, since each customer buys only a tiny part of everything on offer. I also understood why the train-test split has to follow time. It felt natural to split the data randomly, but that would let the model "see" the future, and the results would look better than they really are.

The most interesting part for me was comparing PCA and Truncated SVD. I went in expecting one of them to clearly win, and I was a little surprised when they gave almost the same results. Both did far better than just recommending the best-selling products. SVD was faster for me, and PCA scored very slightly higher, but the difference was so small that I couldn't honestly say either was better. It made me think about how easy it is to pick a "winner" from a tiny gap and present it as a finding. I'm now more careful about that.

The customer segmentation taught me something different. I was happy at first when the groups made sense, like the customers who mostly buy bags or the ones who mostly buy baking items. But then I saw that the groups overlap a lot and aren't clearly separated. That was a good lesson for me. An algorithm will always give you clusters, but that doesn't mean they are real or useful. I now see them as a starting point for exploring, not something I'd present as proven customer types.

I also realise my evaluation has limits. I used the same data to compare settings and to report results, so my numbers are probably a bit optimistic. If I did this again, I would use a separate validation set to choose the settings and keep a final test set that I touch only once. I would also check whether the groups stay the same when I rerun the model, and whether a business could really use them, for example for targeted offers.

Overall, this project helped me move from just running methods to thinking about whether I can trust the results. That is probably the biggest thing I'm taking away from it.